

Publication List

Peter B. Denton

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Articles (79)

- [1] P. B. Denton, S.-F. Ge, C.-F. Kong, and P. Pasquini, “RG Running of Multiple Neutrino Mixing Parameters at Oscillation Experiments,”
[arXiv:2606.26720 \[hep-ph\]](#).
- [2] P. B. Denton, J. Gehrlein, and H. Truelson, “The Future of Lepton Flavor,”
[arXiv:2606.05291 \[hep-ph\]](#).
- [3] J. Banovetz, C.-A. Hebert, P. B. Denton, D. Scolnic, A. Slosar, and C. Walter, “Uncovering the Next Galactic Supernova with the Vera C. Rubin Observatory,”
The Open Journal of Astrophysics (1, 2026) , [arXiv:2601.12094 \[astro-ph.IM\]](#).

*For the latest version see: [peterdenton.github.io](#)

[†]Most author lists are in alphabetical order as that is the standard in particle physics.

- [4] P. B. Denton and S. J. Parke, “Efficient atmospheric, solar, and supernova neutrino propagation through the Earth,” *Phys. Rev. D* **113** no. 5, (2026) 053002, [arXiv:2511.04735 \[hep-ph\]](#).
- [5] L. A. Anchordoqui *et al.*, “Letter of Intent: The Forward Physics Facility,” [arXiv:2510.26260 \[hep-ex\]](#).
- [6] H. Davoudiasl, P. B. Denton, and A. M. Suliga, “Glimpsing Physics of Nano-Hz Gravitational Waves in Neutrinos from Core-Collapse Supernovae,” [arXiv:2510.23713 \[hep-ph\]](#).
- [7] A. Abada *et al.*, “Neutrino Theory in the Precision Era,” [arXiv:2504.00014 \[hep-ph\]](#).
- [8] P. B. Denton and C. Gourley, “Determining the density of the sun with neutrinos,” *Phys. Lett. B* **866** (2025) 139560, [arXiv:2502.17546 \[hep-ph\]](#).
- [9] P. B. Denton, J. Gehrlein, and C.-F. Kong, “Testing new physics in oscillations at a neutrino factory,” *Nucl. Phys. B* **1018** (2025) 117040, [arXiv:2502.14027 \[hep-ph\]](#).
- [10] P. B. Denton, “Neutrino Oscillations in the Three Flavor Paradigm,” [arXiv:2501.08374 \[hep-ph\]](#).
- [11] P. B. Denton and Y. Kini, “Individual neutrino masses from a supernova,” *Phys. Rev. D* **111** no. 10, (2025) 103006, [arXiv:2411.13634 \[hep-ph\]](#).
- [12] H. Davoudiasl and P. B. Denton, “How fast can protons decay?,” *Phys. Rev. D* **111** no. 3, (2025) 035026, [arXiv:2410.19045 \[hep-ph\]](#).
- [13] P. B. Denton, A. Giarnetti, and D. Meloni, “Solar neutrinos and the strongest oscillation constraints on scalar NSI,” *JHEP* **01** (2025) 097, [arXiv:2409.15411 \[hep-ph\]](#).
- [14] P. B. Denton and J. Gehrlein, “A modern look at the oscillation physics case for a neutrino factory,” *Nucl. Phys. B* **1012** (2025) 116818, [arXiv:2407.02572 \[hep-ph\]](#).
- [15] J. F. Acevedo, J. Berger, and P. B. Denton, “Dark matter raining on DUNE and other large volume detectors,” *JHEP* **11** (2024) 011, [arXiv:2407.01670 \[hep-ph\]](#).
- [16] P. B. Denton and S. J. Parke, “Fast and accurate algorithm for calculating long-baseline neutrino oscillation probabilities with matter effects,” *Phys. Rev. D* **110** no. 7, (2024) 073005, [arXiv:2405.02400 \[hep-ph\]](#).
- [17] P. B. Denton and S. J. Parke, “Smallness of matter effects in long-baseline muon neutrino disappearance,” *Phys. Rev. D* **109** no. 5, (2024) 053002, [arXiv:2401.10326 \[hep-ph\]](#).
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- [19] P. B. Denton and J. Gehrlein, “Survey of neutrino flavor predictions and the neutrinoless double beta decay funnel,” *Phys. Rev. D* **109** no. 5, (2024) 055028, [arXiv:2308.09737 \[hep-ph\]](#).
- [20] P. B. Denton and J. Gehrlein, “Neutrino constraints and the ATOMKI X17 anomaly,” *Phys. Rev. D* **108** no. 1, (2023) 015009, [arXiv:2304.09877 \[hep-ph\]](#).
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- [27] A. de Gouvêa *et al.*, “Theory of Neutrino Physics – Snowmass TF11 (aka NF08) Topical Group Report,” [arXiv:2209.07983 \[hep-ph\]](#).
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- [38] D. Caratelli *et al.*, “Low-Energy Physics in Neutrino LArTPCs,” 3, 2022.
[arXiv:2203.00740 \[physics.ins-det\]](#).
- [39] P. B. Denton, “Sterile Neutrino Search with MicroBooNE’s Electron Neutrino Disappearance Data,” *Phys. Rev. Lett.* **129** no. 6, (2022) 061801,
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Conference Proceedings

- [1] **GRAND** Collaboration, R. Alves Batista *et al.*, “The Giant Radio Array for Neutrino Detection (GRAND) Collaboration – Contributions to the 38th International Cosmic Ray Conference (ICRC 2023),” in *38th International Cosmic Ray Conference*. 7, 2023. [arXiv:2308.00120 \[hep-ex\]](#).
- [2] **UHECR** Collaboration, F. Schroeder *et al.*, “Snowmass UHECR Whitepaper: Requirements on Future Instrumentation,” *PoS ICRC2023* (2023) 206.
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- [4] S. J. Parke, P. B. Denton, and H. Minakata, “Analytic Neutrino Oscillation Probabilities in Matter: Revisited,” [arXiv:1801.00752 \[hep-ph\]](#).

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Collaboration Papers

- [1] **DUNE** Collaboration, S. Abbaslu *et al.*, “Reconstruction of atmospheric neutrinos in DUNE’s horizontal-drift far-detector module,” [arXiv:2601.05697 \[hep-ex\]](#).
- [2] **DUNE** Collaboration, S. Abbaslu *et al.*, “Measurement of Exclusive π^+ -argon Interactions Using ProtoDUNE-SP,” [arXiv:2511.13462 \[hep-ex\]](#).
- [3] **DUNE** Collaboration, S. Abbaslu *et al.*, “First Measurement of π^+ -Ar and p -Ar Total Inelastic Cross Sections in the Sub-GeV Energy Regime with ProtoDUNE-SP Data,” [arXiv:2511.11925 \[nucl-ex\]](#).
- [4] **DUNE** Collaboration, S. Abbaslu *et al.*, “Identification of low-energy kaons in the ProtoDUNE-SP detector,” [arXiv:2510.08380 \[hep-ex\]](#).
- [5] **GRAND** Collaboration, J. Álvarez-Muniz *et al.*, “Towards the Giant Radio Array for Neutrino Detection (GRAND): the GRANDProto300 and GRAND@Auger prototypes,” [arXiv:2509.21306 \[astro-ph.IM\]](#).
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- [7] **DUNE** Collaboration, S. Abbaslu *et al.*, “Operation of a Modular 3D-Pixelated Liquid Argon Time-Projection Chamber in a Neutrino Beam,” [arXiv:2509.07012 \[physics.ins-det\]](#).
- [8] **DUNE** Collaboration, S. Abbaslu *et al.*, “Spatial and Temporal Evaluations of the Liquid Argon Purity in ProtoDUNE-SP,” [arXiv:2507.08586 \[physics.ins-det\]](#).
- [9] **DUNE** Collaboration, A. Abed Abud *et al.*, “European Contributions to Fermilab Accelerator Upgrades and Facilities for the DUNE Experiment,” [arXiv:2503.23744 \[physics.acc-ph\]](#).
- [10] **DUNE** Collaboration, A. Abed Abud *et al.*, “DUNE Software and Computing Research and Development,” [arXiv:2503.23743 \[physics.data-an\]](#).
- [11] **DUNE** Collaboration, A. Abed Abud *et al.*, “The DUNE Science Program,” [arXiv:2503.23291 \[hep-ex\]](#).
- [12] **DUNE** Collaboration, A. Abed Abud *et al.*, “The DUNE Phase II Detectors,” [arXiv:2503.23293 \[physics.ins-det\]](#).

- [13] **DUNE** Collaboration, A. Abed Abud *et al.*, “Neutrino interaction vertex reconstruction in DUNE with Pandora deep learning,” *Eur. Phys. J. C* **85** no. 697, (2025) 697, [arXiv:2502.06637 \[hep-ex\]](#).
- [14] **DUNE** Collaboration, A. Abed Abud *et al.*, “The track-length extension fitting algorithm for energy measurement of interacting particles in liquid argon TPCs and its performance with ProtoDUNE-SP data,” *JINST* **20** no. 02, (2025) P02021, [arXiv:2409.18288 \[physics.ins-det\]](#).
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- [16] **GRAND** Collaboration, R. Alves Batista *et al.*, “GRANDlib: A simulation pipeline for the Giant Radio Array for Neutrino Detection (GRAND),” *Comput. Phys. Commun.* **308** (2025) 109461, [arXiv:2408.10926 \[astro-ph.IM\]](#).
- [17] **DUNE** Collaboration, A. Abed Abud *et al.*, “First measurement of the total inelastic cross section of positively charged kaons on argon at energies between 5.0 and 7.5 GeV,” *Phys. Rev. D* **110** no. 9, (2024) 092011, [arXiv:2408.00582 \[hep-ex\]](#).
- [18] **DUNE** Collaboration, A. Abed Abud *et al.*, “Supernova pointing capabilities of DUNE,” *Phys. Rev. D* **111** no. 9, (2025) 092006, [arXiv:2407.10339 \[hep-ex\]](#).
- [19] **DUNE** Collaboration, A. Abed Abud *et al.*, “Performance of a Modular Ton-Scale Pixel-Readout Liquid Argon Time Projection Chamber,” *Instruments* **8** no. 3, (2024) 41, [arXiv:2403.03212 \[physics.ins-det\]](#).
- [20] **DUNE** Collaboration, A. Abed Abud *et al.*, “Doping liquid argon with xenon in ProtoDUNE Single-Phase: effects on scintillation light,” *JINST* **19** no. 08, (2024) P08005, [arXiv:2402.01568 \[physics.ins-det\]](#).
- [21] **DUNE** Collaboration, A. Abed Abud *et al.*, “The DUNE Far Detector Vertical Drift Technology. Technical Design Report,” *JINST* **19** no. 08, (2024) T08004, [arXiv:2312.03130 \[hep-ex\]](#).
- [22] D. Ayzenberg *et al.*, “Fundamental physics opportunities with future ground-based mm/sub-mm VLBI arrays,” *Living Rev. Rel.* **28** no. 1, (2025) 4.
- [23] **DUNE** Collaboration, A. Abed Abud *et al.*, “Impact of cross-section uncertainties on supernova neutrino spectral parameter fitting in the Deep Underground Neutrino Experiment,” *Phys. Rev. D* **107** no. 11, (2023) 112012, [arXiv:2303.17007 \[hep-ex\]](#).
- [24] **DUNE** Collaboration, A. Abed Abud *et al.*, “Highly-parallelized simulation of a pixelated LArTPC on a GPU,” *JINST* **18** no. 04, (2023) P04034, [arXiv:2212.09807 \[physics.comp-ph\]](#).

- [25] **FASER** Collaboration, H. Abreu *et al.*, “Technical Proposal: FASERnu,” [arXiv:2001.03073](#) [[physics.ins-det](#)].
- [26] **FASER** Collaboration, H. Abreu *et al.*, “Detecting and Studying High-Energy Collider Neutrinos with FASER at the LHC,” *Eur. Phys. J.* **C80** no. 1, (2020) 61, [arXiv:1908.02310](#) [[hep-ex](#)].
- [27] **GRAND** Collaboration, J. Álvarez Muñoz *et al.*, “The Giant Radio Array for Neutrino Detection (GRAND): Science and Design,” *Sci. China Phys. Mech. Astron.* **63** no. 1, (2020) 219501, [arXiv:1810.09994](#) [[astro-ph.HE](#)].

Talks (130 including 89 invited)

- [1] “CP Violation from CP Conservation.”. **Invited** chalk talk at Fermilab theory group, June 2026.
- [2] “Nu Theory.”. **Invited** colloquium at University of Tennessee, April 2026.
- [3] “CP-Violation with Neutrino Disappearance and NuFast.” <https://indico.bnl.gov/event/31076/>. **Invited** seminar at Kent State University, April 2026.
- [4] “Nu Theory.” <https://indico.bnl.gov/event/31320/>. **Invited** colloquium at BNL, March 2026.
- [5] “NuFast.” <https://indico.bnl.gov/event/31076/>. Talk at the BNL HET Group, February 2026.
- [6] “CP-Violation with Neutrino Disappearance and NuFast.”. **Invited** seminar at Oxford, UK, January 2026.
- [7] “CP-Violation with Neutrino Disappearance and NuFast.”. **Invited** seminar at Imperial College London, January 2026.
- [8] “CP-Violation with Neutrino Disappearance and NuFast.”. **Invited** seminar at Kings College London, January 2026.
- [9] “Modern Neutrino Oscillation Theory.” <https://indico.cern.ch/event/1568942/contributions/6741263/>. **Invited** plenary talk at NuPhys, Kings College London, January 2026.
- [10] “NuFast.” <https://indico.phys.sinica.edu.tw/event/301/contributions/2540/>. **Invited** plenary talk at the International Joint Workshop on the Standard Model and Beyond, Taipei, December 2025.

- [11] “Neutrino Oscillations in the Three Flavor Paradigm.”
youtube.com/watch?v=OxjUAne79M4.
<https://neutrinos.discussingresearch.com/>. **Invited** pedagogical plenary for
 Neutrinos From Home, December 2025.
- [12] “NuFast.”. **Invited** talk at Stony Brook, November 2025.
- [13] “CP-Violation with Neutrino Disappearance and NuFast.”. **Invited** seminar at
 NYU, November 2025.
- [14] “Solar Neutrinos and the Strongest Oscillation Constraints on Scalar NSI.”. **Invited**
 talk, August 2025.
- [15] “Untangling Systematic Uncertainties from BSM Physics.”
<https://indico.cern.ch/event/1424413/contributions/6468090/>. **Invited**
 plenary at NPN at University of Cincinnati, June 2025.
- [16] “Modern Neutrino Oscillation Theory.”
<https://agenda.hep.wisc.edu/event/2257/contributions/33831/>. **Invited**
 plenary at CIPAN at University of Wisconsin, Madison, June 2025.
- [17] “Towards Solidifying the Discrete Mass Ordering Measurement in the Presence of
 New Physics.”
<https://indico.cern.ch/event/1458241/contributions/6507839/>. **Invited**
 plenary at Magnificent CEvNS, Rio, June 2025.
- [18] “NuFast.”. **Invited** talk at the Stony Brook DUNE group, June 2025.
- [19] “Shining the Neutrino Flashlight into the Darkest Places.”. **Invited** seminar, June
 2025.
- [20] “Testing the Three-Flavor Paradigm.”
<https://indico.global/event/13925/contributions/125625/>. **Invited** plenary
 at the NEAT: Neutrino Experiment and Theory workshop at Colorado State
 University, May 2025.
- [21] “Individual Neutrino Masses From a Supernova.”
<https://indico.cern.ch/event/1488822/contributions/6480082/>. **Invited**
 plenary at the Mitchell Conference at Texas A&M, May 2025.
- [22] “Neutrino Factory.”. **Invited** seminar at Universidad Catolica del Norte, April 2024.
- [23] “Neutrino Factory.” <https://indico.bnl.gov/event/26010/>. Talk at the BNL
 HET Group, March 2025.
- [24] “CP Violation with Neutrino Disappearance.”
<https://indico.cern.ch/event/1454726/contributions/6376522/>. Talk at the
 CERN Neutrino Platform Pheno Week, February 2025.

- [25] “Neutrino Oscillations: Unveiling Mysteries.”. **Invited** colloquium at Virginia Tech, February 2025.
- [26] “Individual Neutrino Masses From a Supernova.”
<https://indico.bnl.gov/event/25821/>. Talk at the BNL HET Group, January 2024.
- [27] “Connecting the Extremes: A Story of Supermassive Black Holes and Ultralight Dark Matter.”
https://physics.ucsd.edu/events/seminars-colloquia/event?event_id=1396. **Invited** seminar at UCSD, October 2024.
- [28] “Dark Matter Raining on DUNE and Other Large Volume Detectors.”
<https://indico.bnl.gov/event/24793/>. Talk at the BNL HET Group, October 2024.
- [29] “Unitarity Violation in Neutrino Physics: Brief Pedagogy.”
<https://indico.fnal.gov/event/66471/>. **Invited** talk at DUNE collaboration call, October 2024.
- [30] “Modern Neutrino Oscillation Theory.”
<https://indico.fnal.gov/event/63406/contributions/297152/>. **Invited** keynote plenary at NuFact, Argonne, September 2024.
- [31] “NuFast: Fast and Accurate Algorithm for Calculating Long-Baseline Neutrino Oscillation Probabilities with Matter Effects.”
<https://agenda.infn.it/event/39753/contributions/240076/>. **Invited** talk at NOW, Otranto Italy, August 2024.
- [32] “CP-Violation with Neutrino Disappearance and NuFast.”
<https://indico.sanfordlab.org/event/69/timetable/#132-talk-dark-matter-raining-o>. **Invited** talk at the Center for Theoretical Underground Physics and Related Areas (CETUP*) workshop, Lead SD, July 2023.
- [33] “NuFast: Fast and Accurate Algorithm for Calculating Long-Baseline Neutrino Oscillation Probabilities with Matter Effects.”
<https://indico.fnal.gov/event/60082/contributions/291412/>. Talk at DUNE Collaboration Meeting at Fermilab, May 2024.
- [34] “CP Violation with Neutrino Disappearance.”. **Invited** seminar at Rice University, April 2024.
- [35] “CP Violation with Neutrino Disappearance.”. **Invited** seminar at Colorado State University, March 2024.
- [36] “CP Violation with Neutrino Disappearance.”
<https://indico.bnl.gov/event/21760/>. Talk at the BNL HET Group, March 2024.

- [37] “CP Violation with Neutrino Disappearance.”
<https://indico.fnal.gov/event/63560/>. Talk at the DUNE LBL WG, March 2024.
- [38] “Knowns and Unknowns in Neutrinos.”
<https://web.mit.edu/lms/news/archives/index.html>. **Invited** colloquium at MIT, Boston, October 2023.
- [39] “LMA-Dark: Large New Physics Effects in Neutrino Oscillations.”. **Invited** talk at the COHERENT collaboration, August 2023.
- [40] “Testing Unitarity of the Leptonic Mixing Matrix with Oscillations: A Focus on Tau Neutrinos.” <https://ifirse.icise.vn/nugroup/nuworkshop2023/program.html>. **Invited** plenary at the 19th Rencontres du Vietnam, July 2023.
- [41] “Light Sterile Neutrinos: A Modern Picture and a Model to Evade Cosmology.”
<https://indico.ihep.ac.cn/event/18269/contributions/135575/>. **Invited** talk at WIN, July 2023.
- [42] “Light Sterile Neutrinos: A Modern Picture and a Model to Evade Cosmology.”
<https://indico.sanfordlab.org/event/53/contributions/822/>. **Invited** talk at the Center for Theoretical Underground Physics and Related Areas (CETUP*) workshop, Lead SD, July 2023.
- [43] “Here Comes the Sun: Solar Parameters in Long-Baseline Accelerator Neutrino Oscillations.”
<https://indico.cern.ch/event/1218225/contributions/5384272/>. Talk at Pheno, Pittsburgh, May 2023.
- [44] “Light Sterile Neutrinos: A Modern Picture and a Model to Evade Cosmology.”
<https://indico.cern.ch/event/1258338/contributions/5307365/>. **Invited** talk at University of Cape Town, April 2023.
- [45] “Knowns and Unknowns in Neutrinos.”. **Invited** colloquium at University of Wisconsin, Madison, April 2023.
- [46] “Neutrinos at Snowmass.”. **Invited** seminar at Kings College London, January 2023.
- [47] “Knowns and Unknowns in Neutrinos.”. **Invited** colloquium at Stony Brook University, New York, October 2022.
- [48] “Light (Fermionic?) Dark Matter.”
<https://indico.cern.ch/event/1189979/contributions/5012521/>. **Invited** talk at the International Conference on Neutrinos and Dark Matter in Egypt, October 2022.
- [49] “Connecting the Extremes: A Story of Supermassive Black Holes and Ultralight Dark Matter.”. **Invited** talk at Dark Matter in Compact Objects, Stars, and in Low Energy Experiments at INT, Seattle, August 2022.

- [50] “Tau Neutrinos: from GeV to EeV.”
<https://indico.fnal.gov/event/22303/contributions/246362/>. **Invited** talk at Snowmass, Seattle, July 2022.
- [51] “Sterile neutrinos at 1 eV.”
https://www.mpi-hd.mpg.de/lin/seminar_theory.en.php. **Invited** seminar at MPI Heidelberg, July 2022.
- [52] “Connecting the Extremes: Story of Supermassive Black Holes and Ultralight Dark Matter.” <https://n3as.berkeley.edu/p/event/su22-jun14/>. **Invited** seminar at UC Berkeley via N3AS, June 2022.
- [53] “Connecting the Extremes: Story of Supermassive Black Holes and Ultralight Dark Matter.” <https://theory.tifr.res.in/~sotu/previous.php>. **Invited** seminar at Tata Institute of Fundamental Research, India, June 2022.
- [54] “Flavor mixing, CP violation, and Unitarity.”
https://neutrino2022.org/program/detail_program. **Invited** plenary talk at Neutrino 2022, Seoul Korea, June 2022.
- [55] “CP Violation at Long-Baseline Neutrino Experiments.”
<https://indico.cern.ch/event/1125426/contributions/4868720/>. **Invited** plenary talk at the Mitchell Conference on Collider, Dark Matter, and Neutrino Physics; Texas A&M, College Station TX, May 2022.
- [56] “Neutrino Theory Overview.”
<https://indico.sanfordlab.org/event/28/contributions/310/>. **Invited** plenary talk at the Conference on Science at the Sanford Underground Research Facility (CoSSURF), May 2022.
- [57] “Nu physics: Theory and practice.”
<https://indico.cern.ch/event/1089132/contributions/4863585/>. **Invited** plenary talk at the Phenomenology Symposium, Pittsburgh, May 2022.
- [58] “CP Violation at Long-Baseline Neutrino Experiments.”
<https://lawphysics.wordpress.com/2022/04/20/w131-peter-denton-cp-violation-at-long-baseline/>. **Invited** seminar in the Latin American Webinar Physics series, April 2022.
- [59] “CP Violation at Long-Baseline Neutrino Experiments.”. **Invited** seminar at Harvard, April 2022.
- [60] “Connecting the Extremes: Story of Supermassive Black Holes and Ultralight Dark Matter.”. **Invited** seminar at MIT, April 2022.
- [61] “New Perspectives on Atmospheric Neutrinos.”. **Invited** seminar at INFN Torino, March 2022.

- [62] “Tau Neutrino Identification at IceCube for Unitary Violation Tests.”
<https://indico.cern.ch/event/1103445/contributions/4724172/>. Talk at Snowmass BSM neutrino workshop, February 2022.
- [63] “Neutrinos and Cosmic Rays at Snowmass.”
<https://indico.bnl.gov/event/13887/>. Talk at BNL Snowmass Retreat, December 2021.
- [64] “Astrophysical Neutrino Decay.”
<https://indico.ipmu.jp/event/397/contributions/6390/>. **Invited** talk at Dark Sectors of Astroparticle Physics at IPMU, Japan December 2021.
- [65] “Neutrino Oscillations at FPF.”
<https://indico.cern.ch/event/1076733/contributions/4577119/>. Talk at Third Forward Physics Facility October 2021.
- [66] “Astrophysical Neutrino Decay.”
<https://indico.ific.uv.es/event/6178/contributions/15526/>. Talk at TAUP August 2021.
- [67] “Astrophysical Neutrino Decay.”
<https://indico.desy.de/event/28202/contributions/105961/>. Talk at EPS-HEP July 2021.
- [68] “CP-Violating Neutrino Non-Standard Interactions in Long-Baseline-Accelerator Data.” <https://indico.cern.ch/event/1034469/contributions/4430079/>. Talk at DPF at FSU July 2021.
- [69] “CP-Violating Neutrino Non-Standard Interactions in Long-Baseline-Accelerator Data.” <https://indico.ibs.re.kr/event/357/timetable/>. Talk at PASCOS in IBS Koreas June 2021.
- [70] “CP-Violating Neutrino Non-Standard Interactions in Long-Baseline-Accelerator Data.” <https://indico.cern.ch/event/982783/contributions/4362341/>. Talk at Pheno May 2021.
- [71] “Neutrino Oscillations in Matter and Linear Algebra.”. **Invited** colloquium at Illinois Institute of Technology April 2021.
- [72] “CP Violation at Long-Baseline Neutrino Experiments.”. **Invited** seminar at Michigan State University March 2021.
- [73] “Astrophysical Neutrino Decay.”
<https://agenda.infn.it/event/24250/contributions/129755/>. Talk at the XIX International Workshop on Neutrino Telescopes February 2021.
- [74] “CP Violation at Long-Baseline Neutrino Experiments.”
<https://indico.cern.ch/event/1001277/>. **Invited** seminar at Sydney CPPC February 2021.

- [75] “Ultralight Fermionic Dark Matter.”. **Invited** talk at Asymptotic Safety and Dark Matter workshop at OSU December 2020.
- [76] “Ultralight Fermionic Dark Matter.” <https://www.ictp-saifr.org/dmw2020/>. Talk at 3rd South American Dark Matter Workshop at ICTP in Sao Paulo December 2020.
- [77] “3+1+NSI and CP Violation.”. **Invited** seminar at KIAS November 2020.
- [78] “CP Violation at Long-Baseline Neutrino Experiments.” <https://indico.bnl.gov/event/8008/>. Talk at BNL HET Group October 2020.
- [79] “CP Violation at Long-Baseline Neutrino Experiments.” <https://npc.fnal.gov/neutrino-seminar-series/>. **Invited** Neutrino Physics Center seminar October 2020 at Fermilab.
- [80] “The Lightest Dark Matter.”. **Invited** seminar October 2020 at University of Sussex.
- [81] “Astrophysical Neutrino Decay.” <https://indico.cern.ch/event/868940/contributions/3899680/>. Talk at ICHEP July 2020 in Prague (virtual).
- [82] “Visible Decay of Astrophysical Neutrinos.” <https://indico.bnl.gov/event/7985/>. Talk at BNL HET Group May 2020.
- [83] “Ultralight Boson Dark Matter Constraints from Superradiance Leveraging the Event Horizon Telescope Collaboration’s Observations of M87*.” <https://indico.cern.ch/event/858682/contributions/3837326/>. Talk at Pheno May 2020 in Pittsburgh, PA (virtual).
- [84] “Beyond the Standard Model physics with accelerator neutrino experiments.” <https://aps-april.onlineeventpro.freeman.com/sessions/15336169/subsession/25117238/>. **Invited** plenary at APS April Meeting 2020 (virtual).
- [85] “LMA-Dark: Large New Physics Effects in Neutrino Oscillations.” <https://indico.bnl.gov/event/7665/>. Talk at BNL HET Group February 2020.
- [86] “Motivation for neutrino precision in oscillations.” <https://indico.bnl.gov/event/7282/>. **Invited** talk at BNL Snowmass Intensity Frontier & Astrophysics Workshop February 2020.
- [87] “Recent results in neutrino oscillation theory.” <https://www.physics.umass.edu/events/2019-11-15-recent-results-neutrino-oscillation>. **Invited** seminar at UMass Amherst November 2019.
- [88] “Realizing the physics goals at DUNE.” <https://indico.fnal.gov/event/21535/other-view>. **Invited** talk at Modules Of Opportunity for DUNE workshop at BNL November 2019.

- [89] “Recent results in neutrino oscillation theory.”
<https://physics.osu.edu/events/high-energy-physics-seminar-peter-dentonbrookhaven->
Invited seminar at OSU November 2019.
- [90] “New physics probes in future neutrino experiments.”
<https://indico.bnl.gov/event/6652/>. **Invited** colloquium at BNL October 2019.
- [91] “Recent results in neutrino oscillation theory.”
<https://indico.cern.ch/event/800930/contributions/3557081/>. Talk at CERN
 Neutrino Platform October 2019.
- [92] “Neutrino theory in the coming years.” <https://indico.bnl.gov/event/6710/>.
Invited talk at BNL Snowmass Discussion October 2019.
- [93] “Recent results in neutrino oscillation theory.”
<https://theory.fnal.gov/events/event/tbd-neutrinos/>. **Invited** theory
 seminar at Fermilab September 2019.
- [94] “Exact neutrino oscillation probabilities in matter.”
<https://indico.ific.uv.es/event/3649/contributions/11349/>. Talk given at
 TomFest at Vanderbilt August 2019.
- [95] “Neutrino oscillation probabilities in matter.”
<https://indico.cern.ch/event/782953/contributions/3444777/>. Talk given at
 the 2019 DPF meeting at Northeastern July 2019.
- [96] “Neutrino self interactions in the early universe.”
<https://indico.cern.ch/event/812851/contributions/3432032/>. **Invited** talk
 at NTN NSI Workshop at Wash U May 2019.
- [97] “Partial neutrino decay resolves icecube’s track and cascade tension.”
<https://indico.bnl.gov/event/5875/>. Talk at BNL HET Group May 2019.
- [98] “Neutrino Oscillation Probabilities in Matter.”
<http://theory.physics.uci.edu/seminars.html>. **Invited** seminar at UC Irvine
 May 2019.
- [99] “Neutrino Oscillation Probabilities in Matter.”
<http://www.theory.caltech.edu/people/carol/seminar.html>. Seminar at
 Caltech May 2019.
- [100] “Partial Neutrino Decay Addresses the Track – Cascade Tension at IceCube.”
<https://indico.cern.ch/event/777988/contributions/3410555/>. Talk at Pheno
 May 2019 in Pittsburgh, PA.
- [101] “Neutrino Oscillation Probabilities in Matter.”
<https://www.phys.psu.edu/seminars/all-seminars>. **Invited** seminar at Penn
 State April 2019.

- [102] “Neutrino Oscillation Probabilities in Matter.”
<https://www.phys.vt.edu/Talks/NeutrinoPhysicsSeminar.html>. **Invited**
 seminar at Virginia Tech February 2019.
- [103] “Analytic and Compact Expressions for Neutrino Oscillations in Matter.”
<https://dx.doi.org/10.5281/zenodo.2642372>. **Invited** talk at PONDD workshop
 at Fermilab December 2018.
- [104] “Finding the Unexpected in IceCube.”. **Invited** N-Talk at Niels Bohr International
 Academy September 2018 in Copenhagen.
- [105] “High Energy Neutrino Parameter Estimation.”. **Invited** talk at GRAND workshop
 at IAP August 2018.
- [106] “New Neutrino Interactions: Breaking Degeneracies and Relaxing Sterile Tensions.”.
Invited seminar at BNL August 2018.
- [107] “Analytic and compact perturbative expressions for neutrino oscillations in matter.”
<https://indico.cern.ch/event/686555/contributions/2977525/>. Talk at the
 International Conference of High Energy Physics (ICHEP) July 2018 in Seoul.
- [108] “Gamma Ray Bursts, Supernovae, Neutrinos, and IceCube.”. **Invited** talk at IIHE
 April 2018 in Brussels.
- [109] “Gamma Ray Bursts, Supernovae, Neutrinos, and IceCube.”. **Invited** talk at DESY
 January 2018 in Zeuthen.
- [110] “Gamma Ray Bursts, Supernovae, Neutrinos, and IceCube.”. **Invited** talk at
 Arizona State University January 2018.
- [111] “Supernova - Gamma Ray Burst - Neutrino Connection.”. **Invited** SUPER-STARS
 talk at DARK Cosmology Center November 2017 in Copenhagen.
- [112] “Gamma Ray Bursts, Supernovae, Neutrinos, and IceCube.”. **Invited** N-Talk at
 Niels Bohr International Academy November 2017 in Copenhagen.
- [113] “Analytic and compact perturbative expressions for neutrino oscillations in matter.”.
Invited seminar at Campinas State University October 2017.
- [114] “COHERENT and the LMA-Dark NSI Solution.”
<https://indico.uu.se/event/324/session/20/contribution/182>. **Invited** talk
 at the Nufact 2017 workshop September 2017 in Uppsala.
- [115] “What We Can Tell About the Sources of IceCube’s Neutrinos, and What IceCube
 Can Tell Us About Gamma Ray Bursts.”
<http://astro.fnal.gov/events/event/tbd-35/>. Astrophysics theory seminar at
 Fermilab August 2017 in Batavia, IL.

- [116] “The Galactic Contribution to IceCube’s Astrophysical Neutrino Flux.”
<https://indico.cern.ch/event/615891/contributions/2608935/>. Talk at TeV Particle Astrophysics at CCAPP in Columbus, OH.
- [117] “Finding Anisotropies in Cosmic Rays and Neutrinos.”
<http://nbia.nbi.ku.dk/nbia-seminars/nbia-seminar-peter-denton/>. **Invited** seminar at the Niels Bohr International Academy astroparticle seminar April 2017 in Copenhagen.
- [118] “Analytic and compact perturbative expressions for neutrino oscillations in matter.”. Talk at the Center of Excellence for Particle Physics at the Terascale at the University of Melbourne December 2016.
- [119] “Spherical Harmonics as a Tool for Finding Anisotropies in UHECR and Astrophysical Neutrino Fluxes.”. **Invited** talk at the Danish Astroparticle Physics Meeting October 2016 in Odense.
- [120] “The Standard Neutrino Oscillation Parameters and a Surprising Alternative Solution.”. **Invited** N-Talk at Niels Bohr International Academy September 2016 in Copenhagen.
- [121] “Analytic and compact perturbative expressions for neutrino oscillations in matter.”
<http://indico.cern.ch/event/432527/contributions/1071859/>. Talk at the International Conference of High Energy Physics (ICHEP) August 2016 in Chicago, IL.
- [122] “Analytic and compact perturbative expressions for neutrino oscillations in matter.”
<http://theory.fnal.gov/seminars/seminars.html>. **Invited** talk at the Fermilab theory seminar July 2016 in Batavia, IL.
- [123] “Methods for Probing New Physics at High Energies.”
<https://events.vanderbilt.edu/index.php?eID=90084>. Successful dissertation defense at Vanderbilt University June 2016 in Nashville, TN.
- [124] “Analytic and compact perturbative expressions for neutrino oscillations in matter.”
<http://www.ccsem.infn.it/issp2016/index.html>. Talk at the International School of Subnuclear Physics May 2016 in Erice, Sicily.
- [125] “Analytic and compact perturbative expressions for neutrino oscillations in matter.”
<https://indico.cern.ch/event/489180/contributions/2158195/>. Talk at Pheno May 2016 in Pittsburgh, PA.
- [126] “Cosmic Ray Anisotropy with Partial Sky Exposure.”. **Invited** seminar November 2015 at CCAPP.
- [127] “The Effect of a Maximum Lepton Energy on the Stability of Pions and Cosmic Ray Physics.” <http://meetings.aps.org/link/BAPS.2015.APR.M14.1>. Talk at the APS April meeting 2015 in Baltimore, MD.

- [128] “Particle Physics at the Highest Energies.”. **Invited** seminar December 2014 at the University of Wisconsin – Madison.
- [129] “Sensitivity of orbiting JEM-EUSO to large-scale cosmic-ray anisotropies.”. Talk at the Cosmic Ray Anisotropy Workshop September 2013 in Madison, WI.
- [130] “Using dispersion relations to look for new physics in pp elastic scattering at the LHC.” <http://meetings.aps.org/link/BAPS.2013.APR.H12.8>. Talk at the APS April meeting 2013 in Denver, CO.

Lectures

- [1] “Neutrino Theory Overview.” <https://indico.global/event/16507/overview>. Lecture at the International Neutrino Summer School at UC Santa Barbara, June 2026.
- [2] “Neutrinos: Atmospheric, Solar, Astrophysical, and More.” <https://npc.fnal.gov/neutrino-university/>. Lecture for the NPC Neutrino University at Fermilab, June 2026.
- [3] “Neutrino Oscillations and Theory Biases.”. Lecture for students at the CETUP workshop in Lead SD, July 2023.
- [4] “Neutrino Oscillations and Theory Biases.” <https://indico.bnl.gov/event/19465/timetable/>. Lecture for students at BNL, June 2023.
- [5] “Neutrino Oscillations and Theory Biases.” <https://indico.bnl.gov/event/15829/timetable/>. Lecture for students at BNL, June 2022.
- [6] “Neutrino Oscillations.”. Two lectures for undergraduates at TIFR, May 2021.

Notes

- [1] P. B. Denton, H. Minakata, and S. J. Parke, “Comment on 1801.10488v3,”. <https://zenodo.org/record/1177535>.

Code

- [1] P. B. Denton and S. J. Parke, “NuFast-Earth.”. <https://github.com/PeterDenton/NuFast-Earth/>.
- [2] P. B. Denton and S. J. Parke, “NuFast-LBL.”. <https://github.com/PeterDenton/NuFast-LBL/>.

- [3] A. Abdullahi and P. B. Denton, “Astro-Nu-Decay.”
<https://github.com/PeterDenton/Astro-Nu-Decay>.
- [4] P. B. Denton, “Peterdenton/nu-pert-compare: v1.0.0,” Jan., 2019.
<https://doi.org/10.5281/zenodo.2547029>.
<https://github.com/PeterDenton/Nu-Pert-Compare>.
- [5] P. B. Denton, “ANA v1.0.0: Astrophysical Neutrino Anisotropy,” Mar., 2017.
<https://doi.org/10.5281/zenodo.438675>.
<https://github.com/PeterDenton/ANA>.
- [6] P. B. Denton, “Nu-Pert v0.2.2: Analytic and compact perturbative expressions for neutrino oscillations in matter,” June, 2016.
<https://doi.org/10.5281/zenodo.54629>.
<https://github.com/PeterDenton/Nu-Pert>.

Miscellaneous

- [1] P. B. Denton* *et al.*, “Neutrino Non-Standard Interactions.” Snowmass 2021: LOI, August, 2020.
https://www.snowmass21.org/docs/files/summaries/NF/SNOWMASS21-NF3_NF1-CF7_CF0-TF11_
 *Editor.
- [2] P. B. Denton* and S. J. Parke, “Direct Probes of the Matter Effect in Neutrino Oscillations.” Snowmass 2021: LOI, August, 2020.
https://www.snowmass21.org/docs/files/summaries/NF/SNOWMASS21-NF1_NF3-TF0_TF0_Peter
 *Editor.
- [3] M. Bustamante*, P. B. Denton*,
 S. Wissel*, *et al.*, “Ultra-High-Energy Neutrinos.” Snowmass 2021: LOI, August, 2020.
https://www.snowmass21.org/docs/files/summaries/NF/SNOWMASS21-NF4_NF6-CF7_CF3-TF9_T
 *Editor.
- [4] P. B. Denton* *et al.*, “Computing Neutrino Oscillations in Matter Efficiently.”
 Snowmass 2021: LOI, July, 2020.
<https://www.snowmass21.org/docs/files/summaries/NF/SNOWMASS21-NF8-CompF2-005.pdf>.
 *Editor.
- [5] L. A. Anchordoqui, M. Bustamante, *et al.*, “Cosmic Neutrino Probes of Fundamental Physics.” Snowmass 2021: LOI, August, 2020.
https://www.snowmass21.org/docs/files/summaries/CF/SNOWMASS21-CF7_CF1-NF4_NF3-TF11_
- [6] L. A. Anchordoqui *et al.*, “Syn-
 ergy of astro-particle physics and collider physics.” Snowmass 2021: LOI, August, 2020.
https://www.snowmass21.org/docs/files/summaries/CF/SNOWMASS21-CF7_CF0-EF6_EF7-NF5_M

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